

Education

- 2026 – 2030

Massachusetts Institute of Technology, Cambridge, Massachusetts

Course 2: Mechanical Engineering Student

- 2022 – 2026

Newark Charter High School, Newark, Delaware

Project Lead The Way (PLTW) – Engineering Pathway

2x Varsity Lacrosse Captain, Physics Club VP

4.42 GPA - weighted, cumulative

- 2023

University of Delaware ENTR167 Intro to Making; Dual Enrollment Course

Technical Experience

Visual Project Portfolio: nhryn.tech

Research Intern - Carpick Research Group, University of Pennsylvania - 2024 - 2026

- **Designed and fabricated** tribometer fixtures that resolved persistent data-reliability issues in grease testing and were adopted lab-wide across multiple unrelated research projects
- Investigated tribofilm growth mechanisms in metal-oxide nanoparticle greases in collaboration with ExxonMobil & Pixelligent, using tribometers, atomic force microscopy, and interferometry
- Designed and fabricated a humidity-controlled enclosure to eliminate capillary-force variability in adhesion testing, now used in contract research for ASML
- Named as 3rd author on a manuscript currently in preparation

Independent Research – Tribology & Additive Manufacturing - 2024 - 2025

- Conducted independent research collaborating with Widener University and the University of Pennsylvania to optimize tribological properties of FDM 3D-printed PETG/PTFE composites, motivated by affordable prosthetics
- **Introduced novel approach** in the field by engineering a self-lubricating composite material rather than optimizing standard filaments, reducing friction 18% versus conventional PETG
- Selected to present research at the 2025 ASME International Mechanical Engineering Congress & Exposition; peer-reviewed proceedings **published:** <https://doi.org/10.1115/IMECE2025-165930>

Technology Student Association - DE State Treasurer - 2022 – Present

- CAD Engineering Competition - 5th in USA
- Software Development Competition - 19th in USA

Jr. Swim Team Coach & Lifeguard - Persimmon Creek Swim Club - 2023

- Worked as Lifeguard and volunteered instructing 4-8 year old swim team members at local pool

Technical Skills

CAD Software - Autodesk Fusion 360, Solidworks, Revit

Programming - Python, MATLAB, Flutter, Firebase

Data - Excel, Gwyddion, ImageJ

Characterization: White-Light Interferometry, Laser Confocal Microscopy, Atomic Force Microscopy